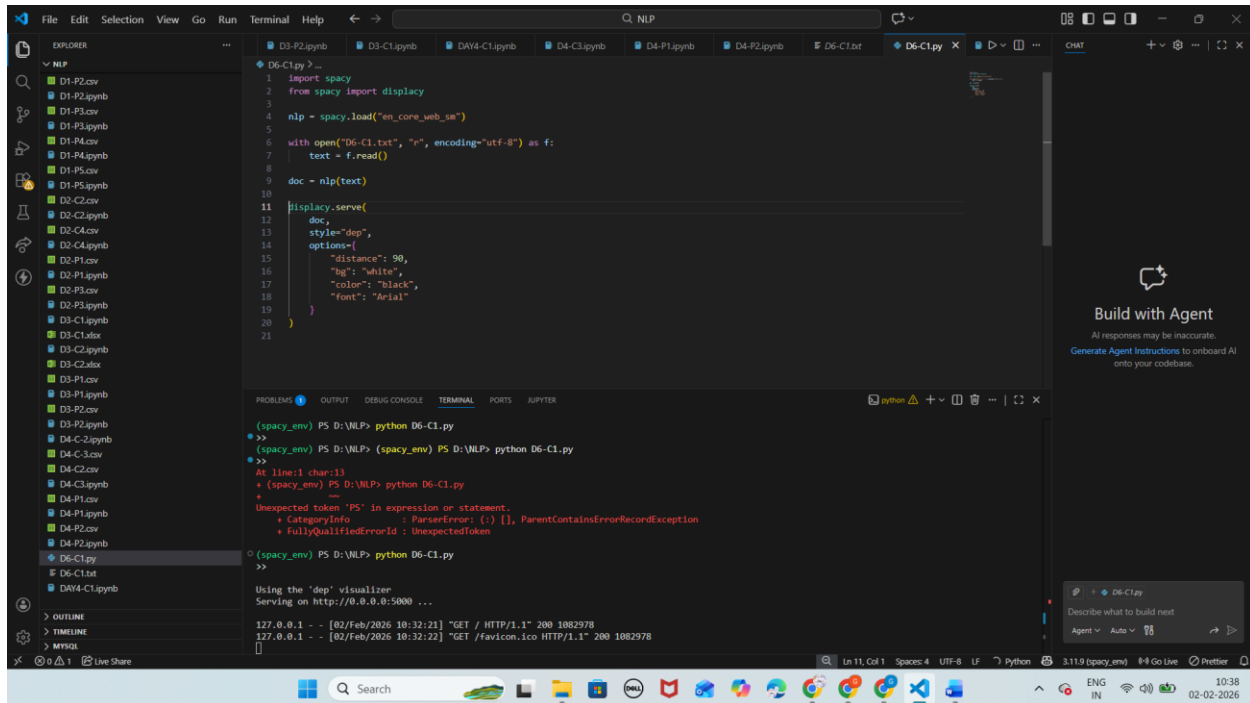


NATURAL PROCESSING LANGUAGE (NLP)

QUESTION -1 DAY – 6 CONCEPT QUESTION

Q.no:1



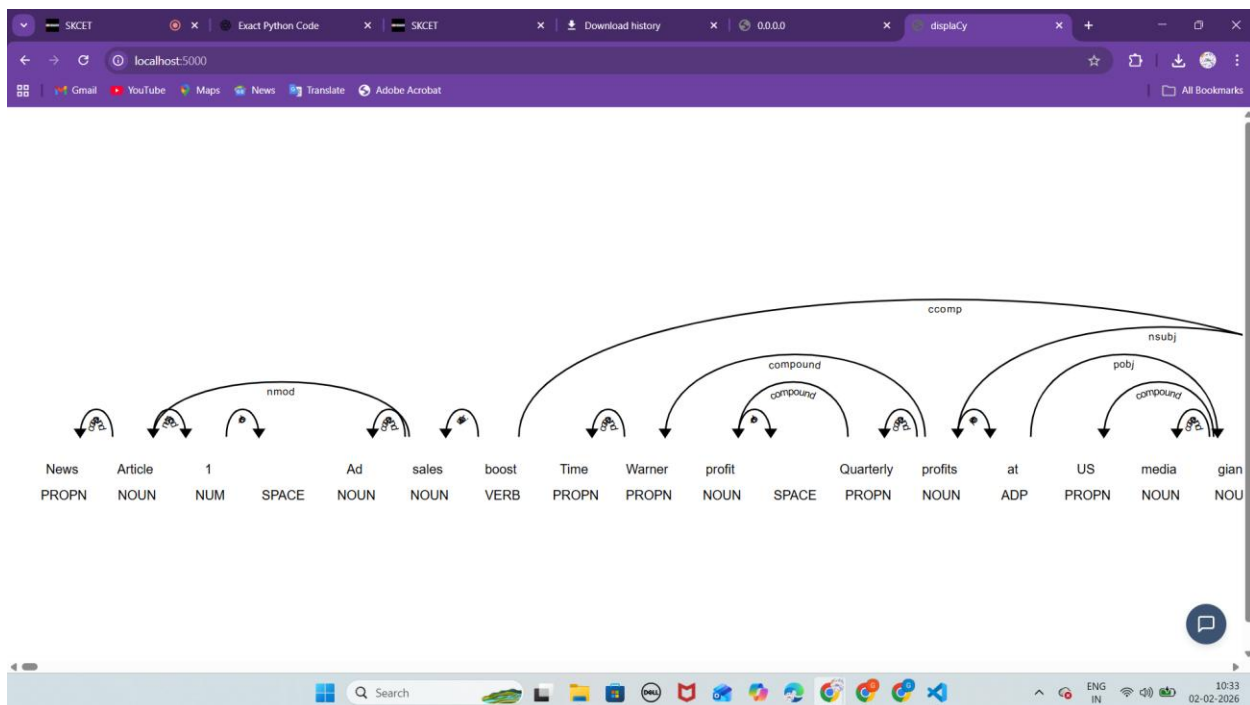
```
1 import spacy
2 from spacy import displacy
3
4 nlp = spacy.load("en_core_web_sm")
5
6 with open("D6-C1.txt", "r", encoding="utf-8") as f:
7     text = f.read()
8
9 doc = nlp(text)
10
11 displacy.serve(
12     doc,
13     style="dep",
14     options={
15         "distance": 90,
16         "bg": "white",
17         "color": "black",
18         "font": "Arial"
19     }
20 )
21
```

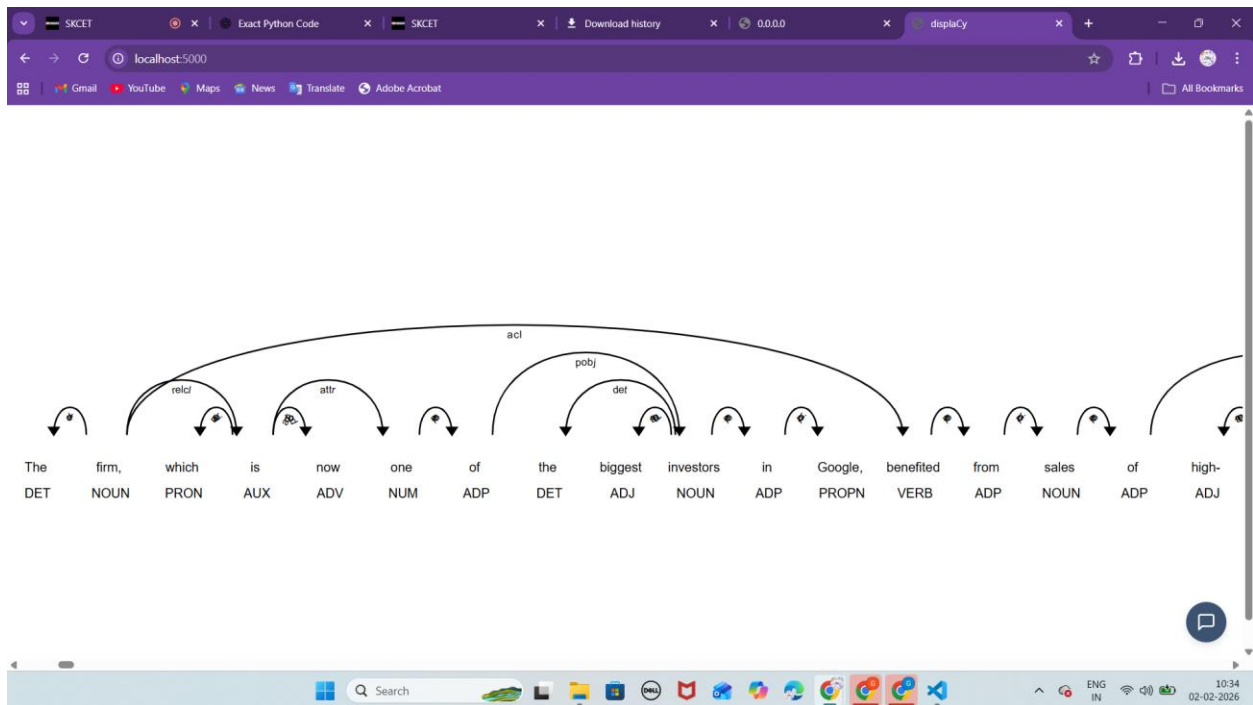
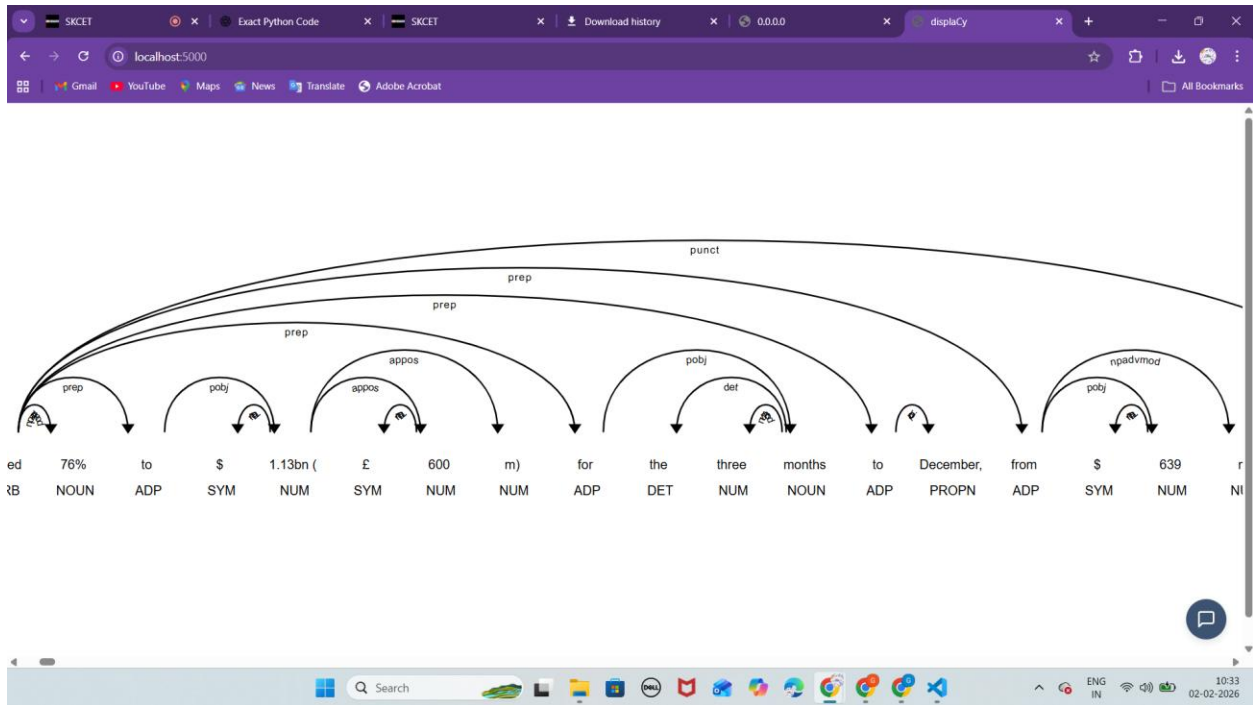
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

```
(spacy_env) PS D:\NLP> python D6-C1.py
(spacy_env) PS D:\NLP> (spacy_env) PS D:\NLP> python D6-C1.py
>>>
At line:1 char:13
+ (spacy_env) PS D:\NLP> python D6-C1.py
~
Unexpected token 'PS' in expression or statement.
+ CategoryInfo          : ParserError: ([]) ParentContainsErrorRecordException
+ FullyQualifiedErrorId : UnexpectedToken

(spacy_env) PS D:\NLP> python D6-C1.py
>>>

Using the 'dep' visualizer
Serving on http://0.0.0.0:5000 ...
127.0.0.1 - - [02/Feb/2026 10:32:21] "GET / HTTP/1.1" 200 1082978
127.0.0.1 - - [02/Feb/2026 10:32:22] "GET /favicon.ico HTTP/1.1" 200 1082978
```





Browser window showing a web application at localhost:5000. The application displays a sentence with its constituent parts and their grammatical roles, connected by arrows indicating syntactic structure.

Sentence: Time Warner said fourth quarter sales rose 2% to \$11.1bn from \$10.9bn. Its profits were buoyed.

Grammatical roles (POS tags):

- Time Warner: PROP
- said: VERB
- fourth: ADJ
- quarter: NOUN
- sales: NOUN
- rose: VERB
- 2%: NOUN
- to: ADP
- \$: SYM
- 11.1bn: NUM
- from: ADP
- \$: SYM
- 10.9bn: NUM
- Its: PRON
- profits: NOUN
- were: AUX
- buoyed: VERB

Arrows indicating syntactic structure:

- Time Warner (PROP) to said (VERB)
- said (VERB) to fourth (ADJ)
- fourth (ADJ) to quarter (NOUN)
- quarter (NOUN) to sales (NOUN)
- sales (NOUN) to rose (VERB)
- rose (VERB) to 2% (NOUN)
- 2% (NOUN) to to (ADP)
- to (ADP) to \$ (SYM)
- \$ (SYM) to 11.1bn (NUM)
- 11.1bn (NUM) to from (ADP)
- from (ADP) to \$ (SYM)
- \$ (SYM) to 10.9bn (NUM)
- 10.9bn (NUM) to Its (PRON)
- Its (PRON) to profits (NOUN)
- profits (NOUN) to were (AUX)
- were (AUX) to buoyed (VERB)

Windows taskbar at the bottom shows the time 10:34 and date 02-02-2026.